

MBAX 6410

Process Analytics

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	None
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus, Online Section Only)	AI Engineer (97.3 / 100)	81.3 / 100

81

CURRENT AVG SCORE
across 6 job roles

100

POTENTIAL SCORE
with recommended additions

+33

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Covers the concepts and tools to design and manage business processes. Emphasizes modeling and analysis, information technology support for process activities, and management of process flows. Graphical simulation software is used to create dynamic models of business processes and predict the effect of changes. Prepares students for a strong management or consulting career path in business processes.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	97.3	100.0	+2.7	LOW
Machine Learning Engineer	96.2	100.0	+3.8	LOW
Data Scientist	78.9	100.0	+21.1	MEDIUM
ML Engineer	75.2	100.0	+24.8	MEDIUM
MLOps Engineer	73.7	97.9	+24.2	MEDIUM
Applied Scientist	66.8	100.0	+33.2	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

LLMs & Generative AI 1 of 6 roles

storage

Deep Learning 1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1

HIGH

Add System Design coverage — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- 2

LOW

Optional: LLMs & Generative AI — would broaden appeal to one additional role. Candidate terms: storage.
- 3

LOW

Optional: Deep Learning — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

Data Scientist

78.9 → 100.0 (+21.1 pts)

rules data sets analytics social build problems approaches

ML Engineer

75.2 → 100.0 (+24.8 pts)

discovery data learning systems model world build problems

MLOps Engineer

73.7 → 97.9 (+24.2 pts)

data model systems learning training development

Applied Scientist

66.8 → 100.0 (+33.2 pts)

biology discovery scientific computational research storage interdisciplinary problems

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.